REPUBLIC OF BOTSWANA

MINISTRY OF ENVIRONMENT,
NATURAL RESOURCES CONSERVATION AND TOURISM

Application for Research Permit

This form should be completed and signed by any applicant who wishes to obtain a permit for conducting research in the Republic of Botswana. The completed form can be submitted by E-mail to mentpermits@gov.bw or post to *Permanent Secretary, Ministry of Environment, Natural Resources Conservation and Tourism, Private Bag BO199, Gaborone, Botswana.* Applicants from foreign countries are advised not to leave for Botswana before obtaining a research permit. A copy of a research project proposal must accompany this application.

1. Title of Research:

Minerals, movement and corridors of bull elephants in the Makgadikgadi: modelling and predicting elephant movement from GPS data to support human-elephant coexistence

2. Name and address of Applicant (include Tel/Fax and E-mail):

Axel Langenskiöld, Vårfrugatan 8, SE-223 50 Lund, Sweden. Tel: +46 73 209 70 84. E-mail: axel@langenskiold.se

3. Name and address of home institutions (if any) to which you are affiliated:

Department of Computer Science, Faculty of Engineering (LTH), Lund University, Box 118, SE-221 00 Lund, Sweden

4. Name and address of supervisor of research (in-home country) or responsible referee:

Jonas Skeppstedt, Department of Computer Science, Lund University, Box 118, SE-221 00 Lund, Sweden. E-mail: jonas.skeppstedt@cs.lth.se. Co-supervisor: Johan Lindström, Mathematical Statistics, Lund University, johan.lindstrom@matstat.lu.se

5. Research plan (attach a research proposal if available):

See attached research proposal. In short: analysis of existing GPS data from bull elephants collected by Elephants for Africa, combined with open environmental data and mineral data from a literature review, to study whether minerals influence where the elephants move and to predict their movements months ahead. If time allows, at most 20 topsoil samples are collected to check the mineral data.

6. Budget for the costs in Botswana (give a detailed breakdown of research costs such as subsistence, travel/transport, local staff, secretarial services, seminar (s), printing, etc.)

Accommodation and meals at Elephants for Africa's research camp, three months of research: BWP 57,700. Research fee, National Park (non-resident), three months: BWP 3,000. Total: BWP 60,700. Optional soil sampling, only if carried out, paid by the applicant: vehicle with driver BWP 3,110 per field day, plus park ferry BWP 400 per day if required, i.e. up to BWP 3,510 per field day. No other costs are planned.

7. Name and address of financial sponsor (s) of the research (if applicable)

Elephants for Africa funds the laboratory analysis of any soil samples. Otherwise the research is self-funded by the applicant.

8. Has funding already been obtained (Yes/No)

Yes.

a) If yes, please state the total amount granted, and the name and address of the funding agency.

Self-funded by the applicant through personal funds and Swedish student finance, covering the budget in question 6 (BWP 60,700, plus up to BWP 3,510 per field day if the optional soil sampling is carried out). Laboratory analysis is funded by Elephants for Africa.

b) If no, what steps are being taken to ensure sufficient funding?

Not applicable.

9. If you have previously done research in Botswana, please give details of the research:

None.

10. Name and address of institution in Botswana to which the researcher is to be affiliated:

Elephants for Africa (Botswana-registered NGO, no. CR12058), PO Box HA148 HAK, Maun, Botswana. Physical address: Leroo La Tau Lodge, Khumaga. Affiliation with a tertiary institution in Botswana is being arranged together with Elephants for Africa; details will follow.

11. Details of Botswana-based personnel that will be involved (names, functions, qualifications):

Stephanie Kong, MSc Biological Science, Director of Elephants for Africa: logistical support at the research camp, and day-to-day contact for questions about the data and the study area. Thatayaone Motsentwa, Associate Degree (Limkokwing University), Research Officer at Elephants for Africa: support with the GPS data, local knowledge of the study area and, if carried out, assistance with the soil sampling. Local academic supervisor: to be confirmed.

12. Places in Botswana where the research is to be undertaken:

Elephants for Africa's research camp north of Khumaga on the Boteti River, within Makgadikgadi Pans National Park. The analysis covers GPS data from the western Makgadikgadi and the Boteti River area. If the optional soil sampling is carried out: the western part of Makgadikgadi Pans National Park along the Boteti River, and community land around Khumaga.

13. Proposed time-schedule for the research:

15 February 2027 - 15 May 2027 in Botswana.

14. Plans for dissemination of research findings:

A written report to the Ministry and the Department of Wildlife and National Parks, and the published master's thesis. All resulting publications will be deposited as required.

15. How are the research findings going to be used in the home country?

The findings form a master's thesis at Lund University, published in Lund University's student paper archive. The methods contribute to research on modelling animal movement.

16. Any other information:

Joint master's thesis project with Gustaf Jensen, who submits a separate application with the same research proposal. If time allows, at most 20 topsoil samples are collected. Botswana lacks the facilities to analyse them, so they would be exported, after an export permit has been obtained. A Supplementary Research Permit from the DWNP will be applied for once this permit is issued. Attached: research proposal, passport copy, CV, details of Botswana-based personnel.

17. Signature of the applicant: ……..

18. Date: ……..

19. FOR OFFICIAL USE ONLY:

Action taken:

Action Officer: …….. **Date:** ……

Permit: Approved/Deferred/Rejected

Notes:

I. Please attach copies of national identity document or passports of proposed research personnel;
II. Attach an up-to-date curriculum vitae;
III. Submit a research proposal;
IV. Applicants from foreign countries are advised not to leave for Botswana before obtaining a research permit from the Ministry.
V. It is a statutory requirement that all resulting publications be deposited directly with the following institutions and any other departments as may be specified in the permit:

1) Director
   National Library Services
   Private Bag 0036
   Gaborone, Botswana

2) Director
   National Archives
   P.O. Box 239
   Gaborone, Botswana

3) User Ministry and/or affiliating body

4) Director
   Research and Development Office
   University of Botswana
   Private Bag 0022
   Gaborone, Botswana

VI. For any correspondence pertaining to a research permit issued, always quote the permit reference number.

VII. Fees payable for research conducted in a National Park or Game Reserve (in BWP):

A. Research activities	**Citizen**	**Resident**	**Non-Resident**
1. Research fee, per person, per month or part thereof. The research fee applies to each individual named on the permit and any support staff and includes park/reserve entry and camping.	250	500	1000
B. Motor Vehicles, aircraft and boats inside a National Park or Game Reserve		**Botswana registered**	**Foreign registered**
2. Motor vehicles under 3500 kg, private, per day		10	50
3. Motor vehicles under 3500 kg, commercial, per day		30	200
4. Motor vehicles under 3500 kg, commercial, annual fee		1,500	N/A
5. Motor vehicles between 3500 and 7000 kg, per day		500	1,000
6. Motor vehicles between 3500 and 7000 kg, supply/utility, annual fee		3,000	N/A
7. Motor vehicles more than 7000 kg, per day		800	1,500
8. Aircraft, per day		50	250
9. Aircraft, annual fee		1,500	3,000
10. Boat, power, per day		10	N/A
11. Boat, power, annual fee		1,500	N/A

- These fees do not apply to the Gemsbok National Park or the educational reserves in Gaborone, Francistown and Maun.
- Further details or clarification on fees can be obtained from Parks and Reserves Reservations Office (PARRO) on Tel: +2673996543; E-mail: dwnp@gov.bw.